

THE CALLING VR

A Musical Virtual Reality Experience

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ABSTRACT

In the three plus years since the COVID-19 pandemic shut down Broadway and regional theaters across the United States, many performing arts institutions are slowly working their way back to pre-pandemic levels in terms of annual revenue [Weinert, 2023]. During this period, many theaters turned to sliding scale ticket models [Mangi, 2022] and hybrid in-person/streamed theater seasons to combat increased inflation related production costs [Vincentelli, 2021].

As we use the lessons learned from the pandemic shutdown to shape the future of American theater, there is a golden opportunity for musical theater producers to embrace virtual reality (VR) technology to reduce development and production related expenses ahead of staging their next musical for a live audience.

The Calling: A Musical VR Experience serves as an immersive historical fantasy inspired by the events that transpired during the infamous Memphis Sanitation Strike of 1968. Created in Unity for Oculus Quest I, II, and The Rift S, The Calling was developed as a “guided experience” serving as the prologue for a larger historical musical currently under development with the Tony Award winning producing team at Apples & Oranges Studios.

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1 INTRODUCTION

A historical fantasy inspired by the events that transpired during the infamous Memphis Sanitation Strike, The Calling features custom character and world design with an all-original gospel-infused score and songs. Accessible via Oculus Quest I, II and The Rift S, The Calling combines passive viewing with immersive storytelling and audience interaction to transport you from present day Beale Street back to March 28, 1968 as you join Dr. Martin Luther King Jr. for

his final peaceful demonstration and discover an ancient ancestral power lying dormant within.

Designed as a companion piece to the full-length musical, *At the River I Stand* by Alani iLongwe and Ren Casey, The Calling is Apples and Oranges’ second virtual reality musical production following the live VR experience *Winter Lights* created in partnership with Microsoft and The Discovery Cube Orange County. Best known for producing Broadway musicals *An American in Paris*, *Memphis* and *Hair*, Apples and Oranges leverages new technology to develop, produce and distribute impactful stories in innovative ways.

2 TECHNICAL DESIGN & DEVELOPMENT

The Calling is a virtual reality experience developed for both PC tethered and stand-alone operation, the Oculus Rift S and Meta Quest 2 respectively. The Meta Quest 2 was chosen as a platform for distribution due to its ease of use and consumer availability. This stand-alone head-mounted display also affords obstruction free movement within the three-dimensional world and scenes of The Calling.

The experience was developed using the Unity 3D game engine. A primary theme of The Calling is to transport viewers in time and space to the historical moments and locations of the 1968 Sanitation Workers strike. Unity provides the physics, graphical and audio tools to create a media rich spatial experience that allows viewers to move within these spaces. As a participatory narrative, movement and interaction are key to discovering the story. Unity’s XR Interaction Toolkit framework was used as a hardware agnostic in/out layer between the programmed interactions of the application and the hardware controllers and head-mounted display of the Quest 2 and Rift S.

Movement was configured to be continuous (as opposed to teleporting, which is a common movement method in virtual reality) with a set 45 degree snap rotation that allowed viewers to roam the spaces in a natural motion. This was essential when combined with the spatial audio and narrative evolution of the experience that required matching location in a linear fashion to experience dialogue and music that serve to move the story forward. Touch interaction was used to activate narrative vignettes (moments) of the experience that provide essential detail about the historical events and places. Two buttons on the controllers, the trigger and A button, are used to bring up an exit menu and to confirm choices and progress specific on-screen prompts. The minimal button interactions encouraged viewers to be immersed in the world while also providing utility and comfort should they experience dizziness

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due to the movement or the display, which is common for first time virtual reality users.

Spatial audio is an essential component of *The Calling*. In the experience it is used for immersive staging in the case of ambient sound effects that create a convincing location, and for narrative progression in the case of dialog and the musical elements of the calling itself. In development, Unity's built-in spatial audio components were used to place 3D sound objects and clips that provided unique localization for elements of *The Calling* related to specific environmental features and people key to the narrative. Audio sources were also used in 2D mode to afford communication of principle dialog in large spaces where the free roaming nature of the viewer movement might place them away from the characters. Some audio sources utilized a spatial blend where the sound would transition from fully 3D to fully 2D based on viewer distance to the source using a custom spatial blend curve. This affords localization of the source from a distance to match the expectations of authentic sound perception (3D blend) then transitioning to balanced stereo propagation (2D blend) that immerses the viewer in the audio.

3 SET UP & INSTALLATION

A 10x10 foot space with one chair per head set is required for the installation. *TheCallingVR.exe* should be pre-installed on all headsets and users should follow the instructions below to launch and play the experience.

- Double click on *The CallingVR.exe* to begin the experience
- The experience is set to Room Scale by default. To set the experience to Seated mode, while in the opening screen, press the flat Oculus Home button on the right controller

and select Reset View, and while seated click the trigger to reset the view.

- The left stick controls forward and backward movement
- The right stick turns to the left or right in 45 degree increments
- The X button on the Left controller activates the Quit menu
- If at any time you feel dizzy, please close your eyes for a moment. If the feeling continues, please remove the headset.

4 DISCUSSION

In developing *The Calling* during the pandemic shut down, VR offered Apples & Oranges Studios a unique opportunity to curb the average \$8MM-\$15MM Broadway musical production costs by running virtual rehearsals and building virtual assets that will be used in the live production.

Additionally, *The Calling* serves as a stand-alone experience that, when coupled with dramaturgical materials, can aid educators in teaching high school students about an often-forgotten chapter in the Civil Rights Movement while providing inspiration for how VR technology can be utilized to bridge the gap between technology and musical theater audiences around the world.

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